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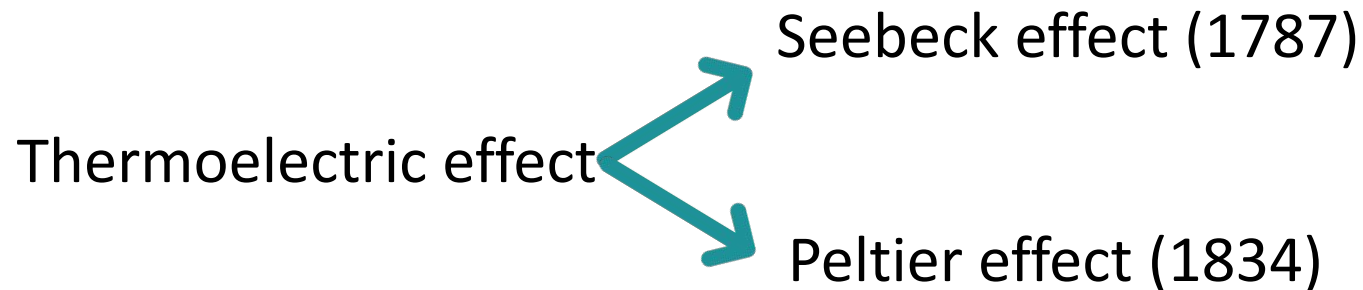
Electron transport properties

ELEONORA ASCRIZZI - MSSC2025 – 12/09/2025

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Thermoelectricity

In all materials **heat is converted into electricity** and viceversa, but in most cases, this phenomenon is negligible. Significant thermoelectric effects are observed only in few **thermoelectric materials**.

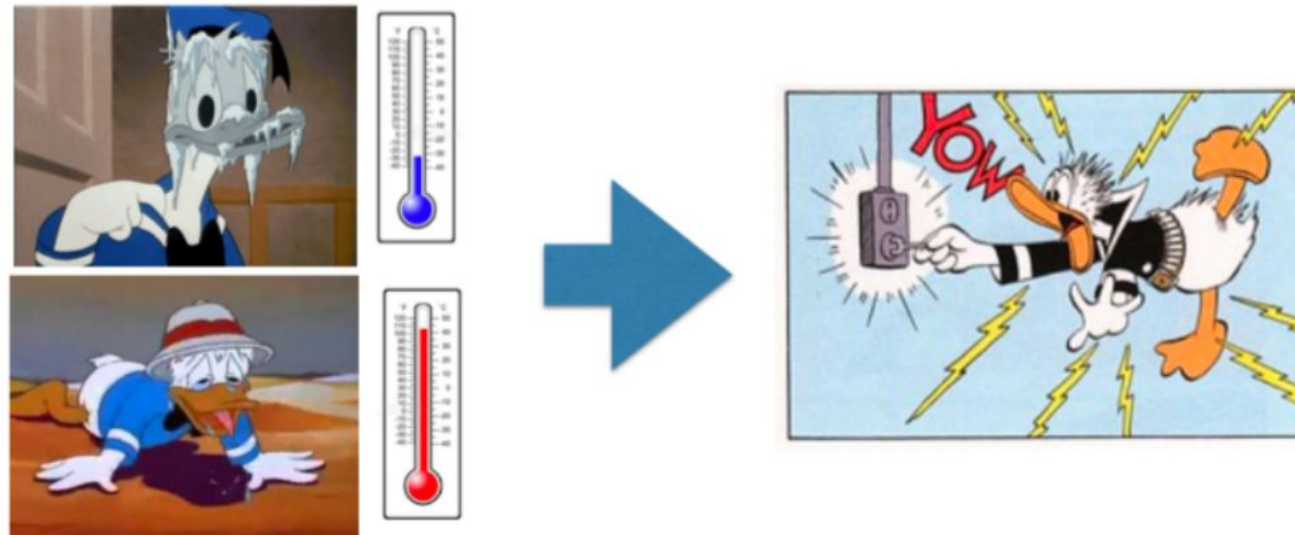


Seebeck Effect

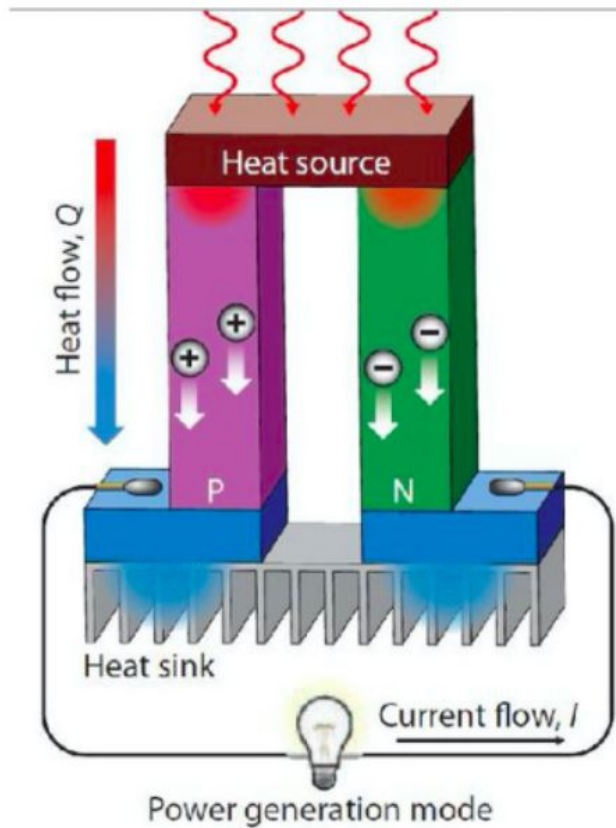
The thermoelectric effect of a material allows to turn a **temperature gradient** into **electric potential**

$$\Delta V = S \Delta T$$

S=Seebeck coefficient

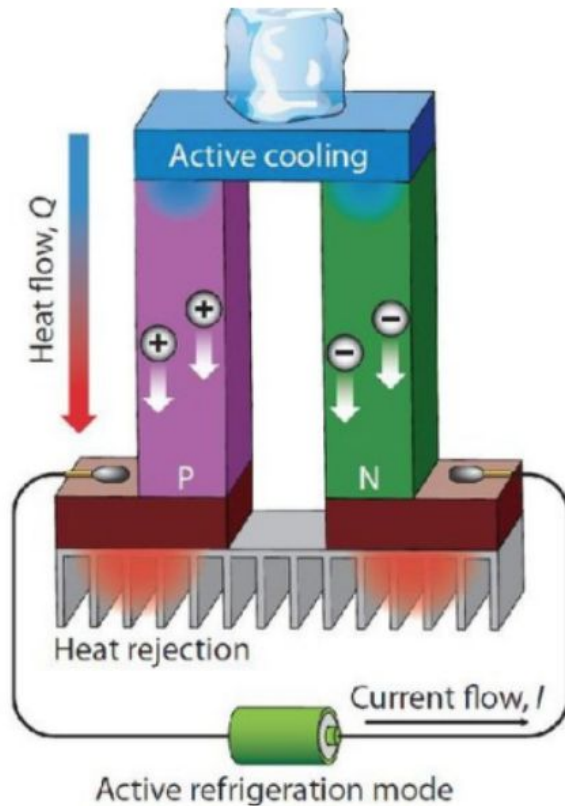


Seebeck Effect



A current is generated from a temperature gradient

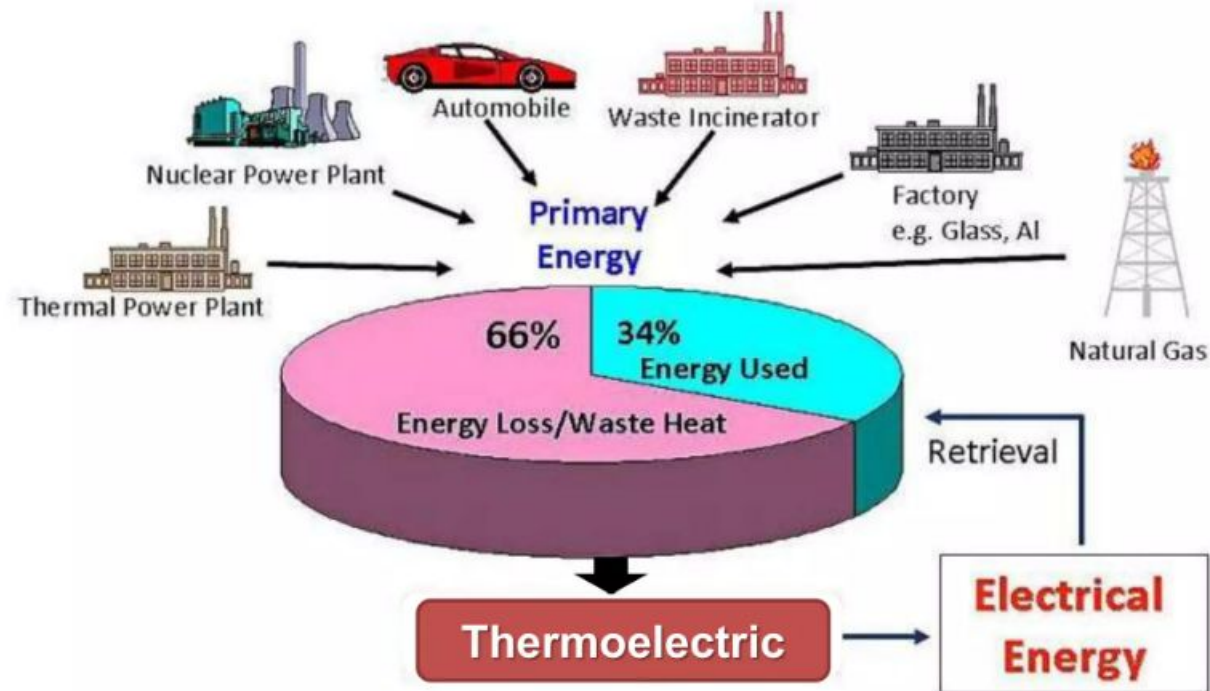
Peltier Effect



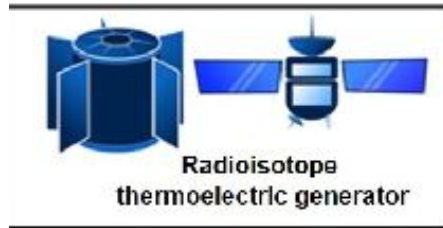
A temperature gradient is generated from a voltage. This is the opposite of the thermoelectric effect and is used in cooling systems

Thermoelectric Materials: applications

Waste heat recover



Thermoelectric Materials: applications



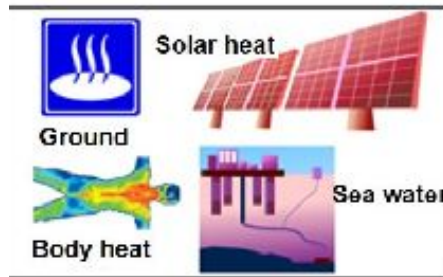
Space Exploration

Good way to recharge devices and prolong their life



Automotive

Collect waste heat to recharge electric vehicles



Wearable devices

Collect body heat to recharge phones, smartwatches...

The Onsager and Callen Model

These equations can provide expression of the three **electron transport coefficients**: electrical conductivity, Seebeck coefficient, electronic contribution to the thermal conductivity

Current density	Transport Coefficients	Forces	$J_E =$ electrical current density $J_Q =$ heat current density $\sigma =$ electrical conductivity $S =$ Seebeck coefficient $T =$ temperature $V =$ electric potential $k_{el} =$ electronic contribution to thermal conductivity
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$$\begin{bmatrix} J_E \\ J_Q \end{bmatrix} = \begin{bmatrix} \sigma & \sigma S \\ T \sigma S & k_{el} \end{bmatrix} \begin{bmatrix} -\nabla V \\ -\nabla T \end{bmatrix}$$

The Seebeck Coefficient

From the Onsager and Callen Model:

$$\mathbf{J}_E = \sigma (-\nabla V - S \nabla T)$$

If $\mathbf{J}_E = 0$

$$\nabla V = -S \nabla T$$

Historically this is a classic example of electromotive force = $-S \nabla T$

The Seebeck Coefficient

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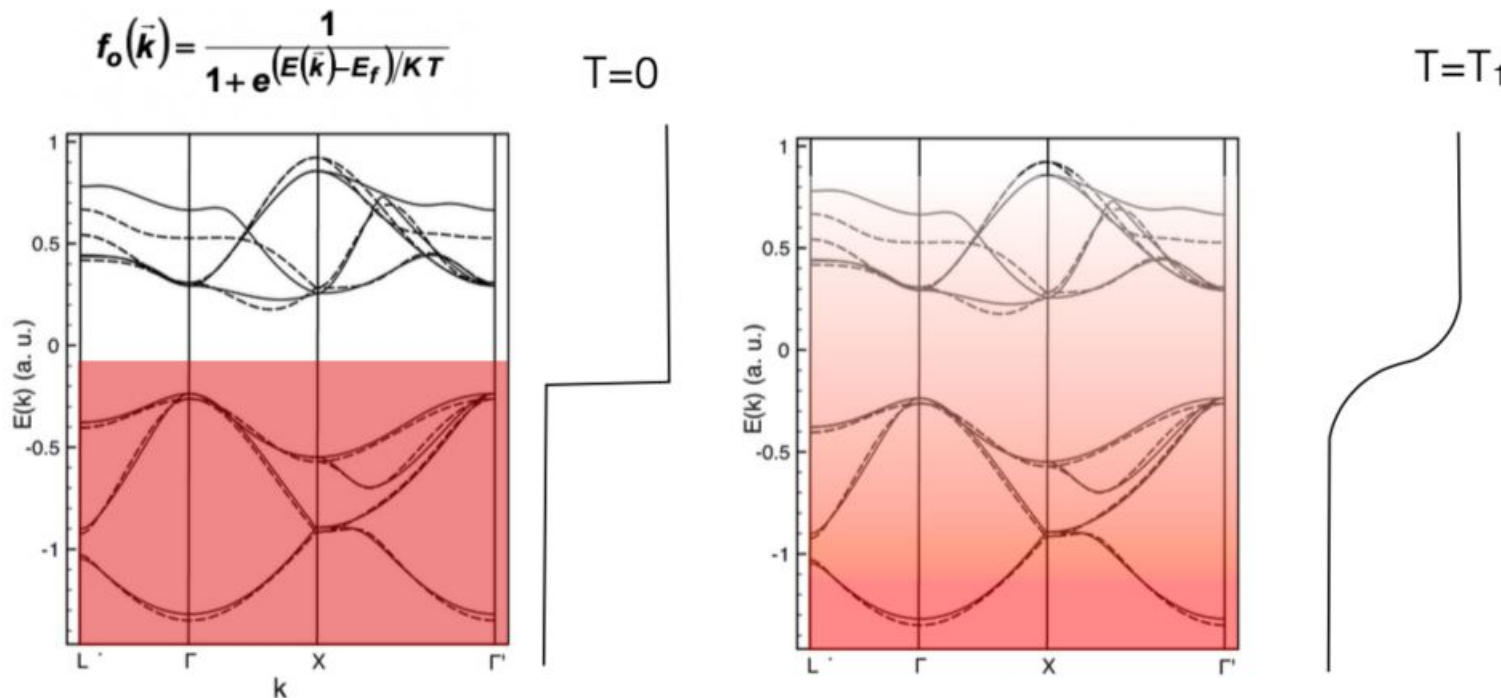
Physical laws behind this phenomenon are known, but modeling them is not an easy task!

Electron transport properties in CRYSTAL

A semiclassical model is considered, the Boltzmann Transport Equation (**BTE**) equation is solved considering three main approximations:

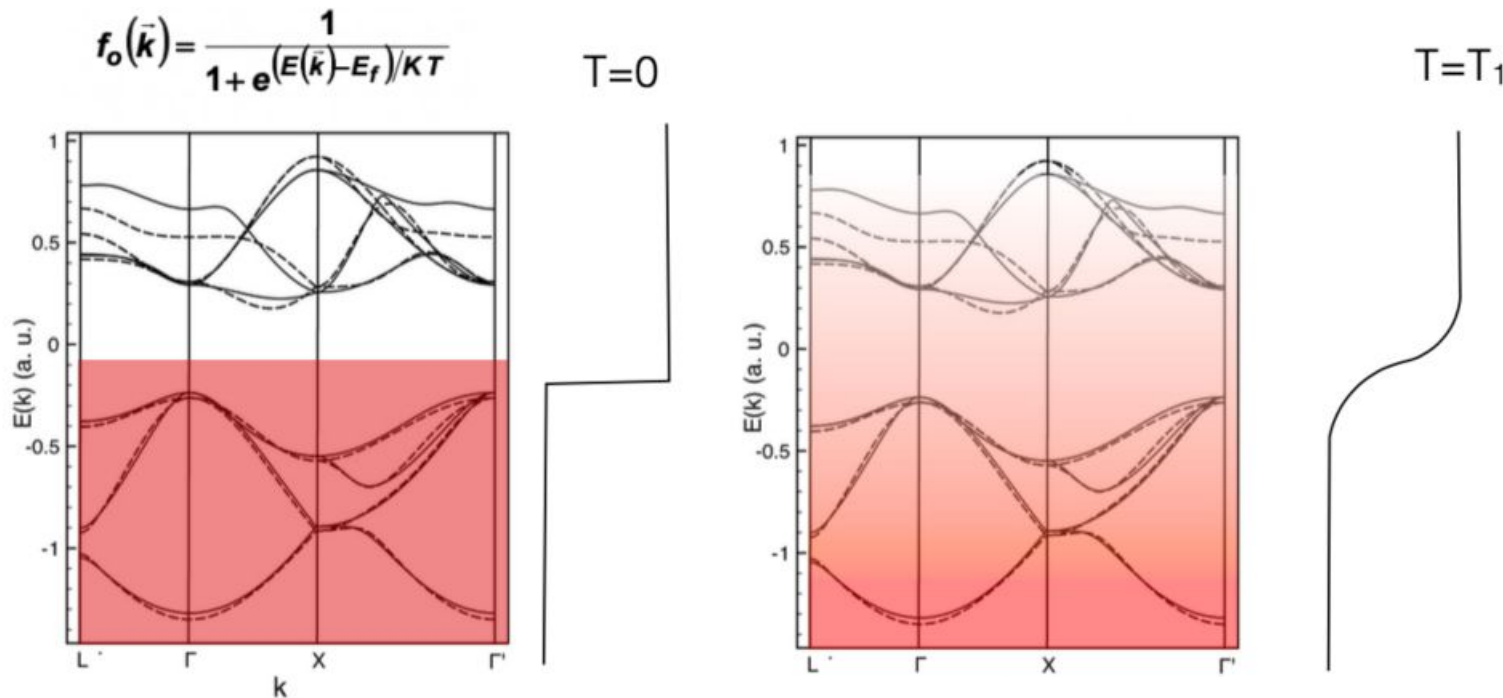
1. A Fermi-Dirac occupation of bands
2. Frozen band approximation
3. Constant relaxation time approximation

1. Fermi-Dirac occupation of bands



Electrons are described by a **Fermi-Dirac distribution function** which is constant in time, i.e. it is always an equilibrium distribution function

2. Frozen bands



The band structures considered is always the same computed at 0K (**frozen band** approximation).

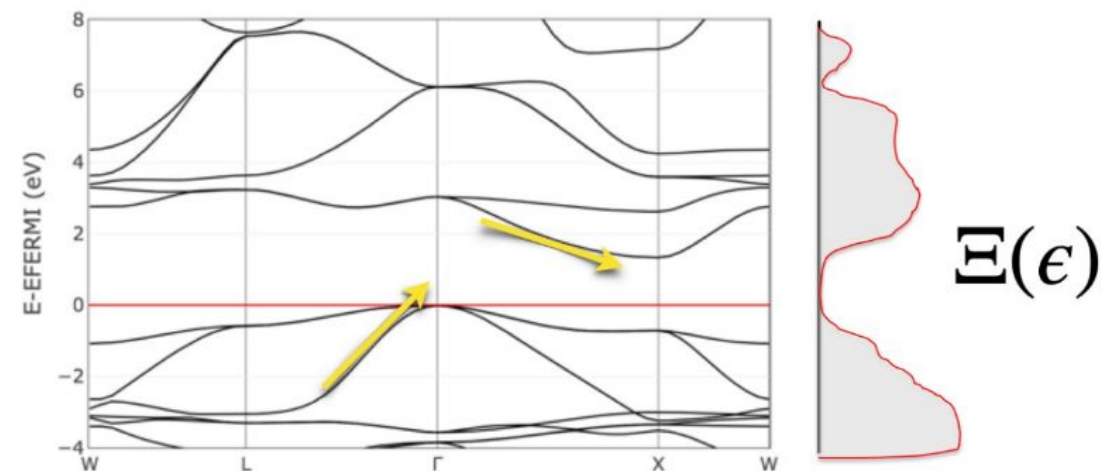
The aim of the model is to relate the band structure to the transport properties

Band structure

$$E_i(\mathbf{k})$$

Band velocity

$$v_{i,q}(\mathbf{k}) \quad \frac{\partial E_i(\mathbf{k})}{\partial k_q}$$



$$\frac{\partial E_i(\mathbf{k})}{\partial k_q} = \left[\mathbf{C}^\dagger(\mathbf{k}) \frac{\partial \mathbf{F}(\mathbf{k})}{\partial k_q} \mathbf{C}(\mathbf{k}) \right]_{ii} - \left[\mathbf{C}^\dagger(\mathbf{k}) \frac{\partial \mathbf{S}(\mathbf{k})}{\partial k_q} \mathbf{C}(\mathbf{k}) \mathbf{E}(\mathbf{k}) \right]_{ii} = v_{i,q}(\mathbf{k})$$

$$\frac{\partial \mathbf{F}(\mathbf{k})}{\partial k_q} = \sum_{\mathbf{g}} i g_q \mathbf{F}(\mathbf{g}) e^{i\mathbf{k} \cdot \mathbf{g}}$$

$$\frac{\partial \mathbf{S}(\mathbf{k})}{\partial k_q} = \sum_{\mathbf{g}} i g_q \mathbf{S}(\mathbf{g}) e^{i\mathbf{k} \cdot \mathbf{g}}$$

3. Constant relaxation time approximation

Transport Distribution Function

$$\Xi_{qr} = \sum_n \sum_{\mathbf{k}} \tau_n [\epsilon_n(\mathbf{k})] \mathbf{v}_{q,n}(\mathbf{k}) \mathbf{v}_{r,n}(\mathbf{k})$$

$$\Xi_{qr} = \tau \sum_n \sum_{\mathbf{k}} \mathbf{v}_{q,n}(\mathbf{k}) \mathbf{v}_{r,n}(\mathbf{k})$$

Constant relaxation time approximation: the relaxation time (τ) is the time that intercourses between collisions that electrons experience.

Ideally it depends on the position of the electron and on the scattering mechanism. In CRYSTAL it is considered to be constant

3. Constant relaxation time approximation

Transport Distribution Function

$$\Xi_{qr} = \sum_n \sum_{\mathbf{k}} \tau_n [\epsilon_n(\mathbf{k})] \mathbf{v}_{q,n}(\mathbf{k}) \mathbf{v}_{r,n}(\mathbf{k})$$

$$\Xi_{qr} = \tau \sum_n \sum_{\mathbf{k}} \mathbf{v}_{q,n}(\mathbf{k}) \mathbf{v}_{r,n}(\mathbf{k})$$

That's everything we need to compute transport coefficients!

Constant relaxation time approximation: the relaxation time (τ) is the time that intercourses between collisions that electrons experience.

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Transport Coefficients

Solving the BTE in a semi-classical model:

$$\sigma(\mu, T) = \int \left(-\frac{\partial f}{\partial \epsilon} \right) \Xi(\epsilon) d\epsilon$$

Electron conductivity

$$\sigma \mathbf{S}(\mu, T) = \frac{1}{T} \int \left(-\frac{\partial f}{\partial \epsilon} \right) (\epsilon - \mu) \Xi(\epsilon) d\epsilon$$

Seebeck coefficient

$$k_e(\mu, T) = \frac{1}{T} \int \left(-\frac{\partial f}{\partial \epsilon} \right) (\epsilon - \mu)^2 \Xi(\epsilon) d\epsilon$$

Electron thermal conductivity

Experimental
Comparison!

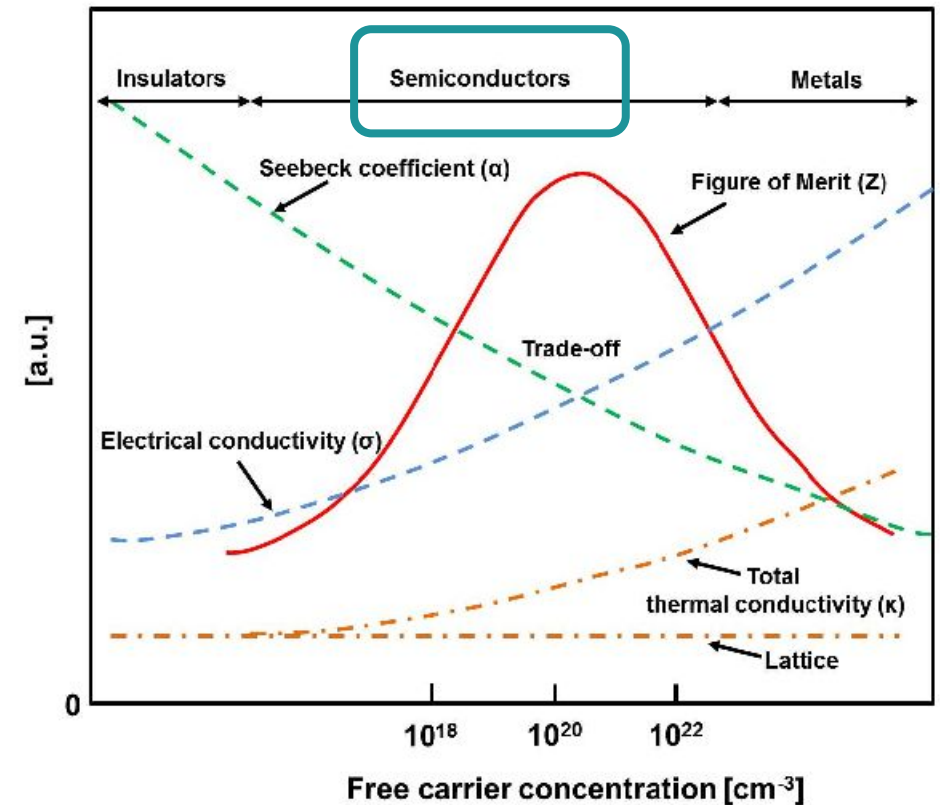
μ = chemical potential, f = Fermi distribution function, ϵ = energy, Ξ = transport distribution function

Thermoelectric Figure of Merit ZT

ZT depends on all the transport coefficient of a material and it is used to compare the overall efficiency of different thermoelectric materials.

$$zT = \frac{\sigma S^2 T}{\kappa}$$

The ideal thermoelectric material has high S , high σ and low κ . To have a sufficient conversion efficiency ZT should be greater than 1 but this is a formidable challenge.



Thermoelectric Figure of Merit ZT

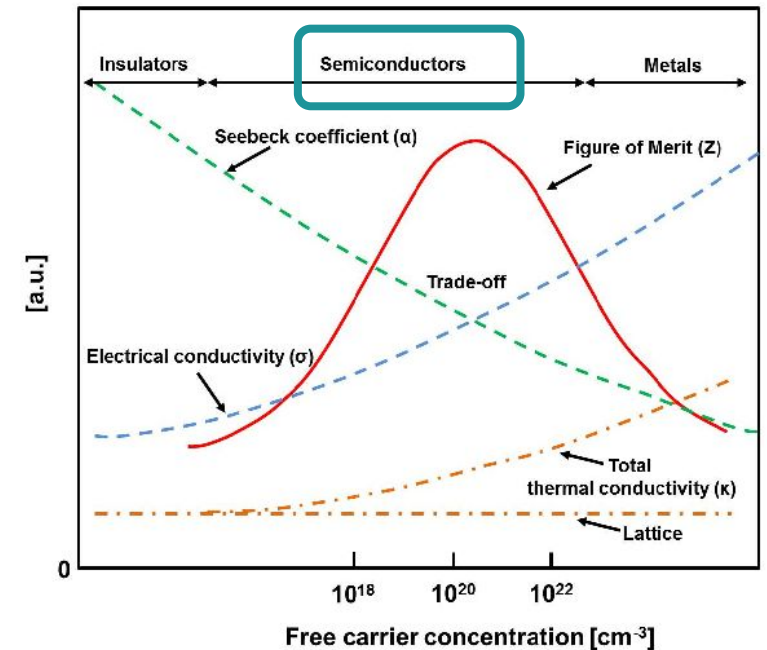
ZT depends on all the transport coefficient of a material and it is used to compare the overall efficiency of different thermoelectric materials.

$$zT = \frac{\sigma S^2 T}{\kappa}$$

κ total thermal conductivity

$$\kappa = \kappa_e + \kappa_p$$

→ Cannot be computed in CRYSTAL at the moment, we rely on experimental data



Electron Transport Properties in CRYSTAL

CRYSTAL performs **directly** the thermoelectric properties¹ calculation, systems of different periodicity can be considered and treated with hybrid functionals.

No post process of DFT wavefunction is needed like in BoltzTrap² or BoltzWann³

The semi-classical Boltzmann Theory is considered, with some limitations:

1. Frozen band approximation
2. Relaxation time approximation
3. No quantities related to the lattice conductivity K_p (essential for a correct esteem of the figure of merit)

Our works on thermoelectric materials

Half-Heusler Alloys, carbon nanutubes, InGaN alloys

The Journal of Physical Chemistry C > Vol 124/Issue 28 > Article

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C: ENERGY CONVERSION AND STORAGE; ENERGY AND CHARGE TRANSPORT | June 22, 2020

Key Role of Defects in Thermoelectric Performance of TiMSn (M = Ni, Pd, and Pt) Half-Heusler Alloys

Atreyi Dasmahapatra, Loredana Edith Daga, Antti J. Karttunen, Lorenzo Maschio*, and Silvia Casassa*

PAPER

Experimental and computational study of the role of defects and secondary phases on the thermoelectric properties of $\text{TiNi}_{1+x}\text{Sn}$ ($0 \leq x \leq 0.12$) half Heusler compounds

Eleonora Ascrizzi¹, Silvia Casassa¹, Loredana Edith Daga¹, Atreyi Dasmahapatra¹, Lorenzo Maschio¹, Antti J. Karttunen², Stefano Boldrini³, Alberto Ferrario³, Carlo Fanciulli⁴, Francesco Aversano¹ [Show full author list](#)

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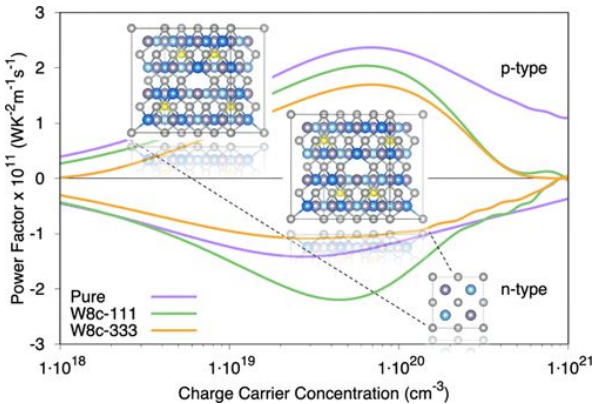
Crucial Role of Ni Point Defects and Sb Doping for Tailoring the Thermoelectric Properties of ZrNiSn Half-Heusler Alloy: An Ab Initio Study

by Eleonora Ascrizzi, Chiara Ribaldone and Silvia Casassa*

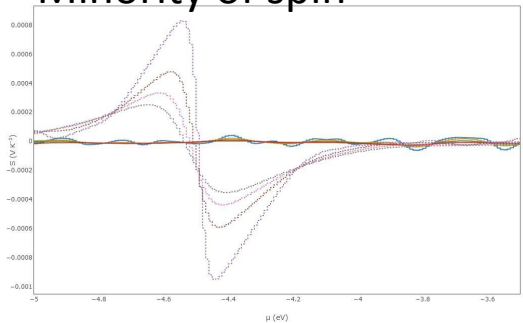
Department of Chemistry, University of Torino, Via Giuria 5, 10125 Torino, Italy

* Author to whom correspondence should be addressed.

Materials **2024**, 17(5), 1061; <https://doi.org/10.3390/ma17051061>



Minority of spin



Modeling of BN-Doped Carbon Nanotube as High-Performance Thermoelectric Materials

by Naiara L. Marana¹, Julio R. Sambrano² and Silvia Casassa^{1,*}

¹ Theoretical Group of Chemistry, Chemistry Department, Torino University, 10125 Torino, Italy

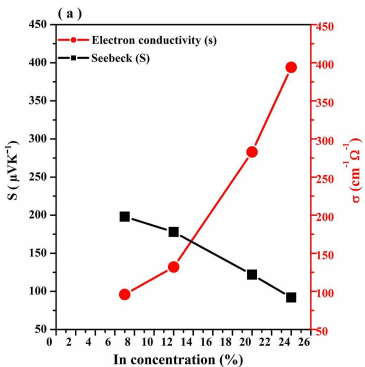
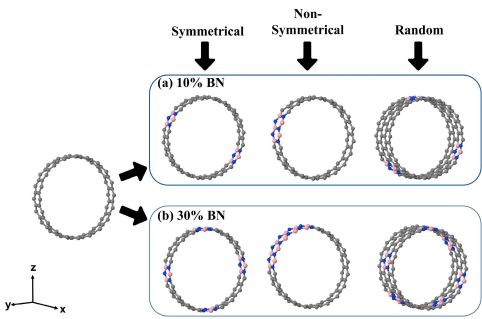
² Modeling and Molecular Simulations Group, São Paulo State University, UNESP, Bauru 15385-000, SP, Brazil

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Nanomaterials **2022**, 12(23), 4343; <https://doi.org/10.3390/nano12234343>

Effect of in content on thermoelectric performance of $\text{In}_x\text{Ga}_{1-x}\text{N}$ alloys: Hybrid density functional theory study based on realistic models

H.C. Aroussi^{a,b}, N.L. Marana^{c,b}, F. Hamdache^a, R. Houaria^{a,d}, S. Bahlouli^e, S. Casassa^b



Majority of spin

In progress: Full Heusler semimetals

Thank you for your attention!